

Using existing AI models vs custom models

For enterprises, there are two choices: building customised AI models and AI agents or using existing models available on platforms like GitHub. Each has its own advantages and disadvantages, but it's up to the developers to choose, keeping in mind the time and resources it will save.

Three factors impact the decision.

- **Custom model training costs:** Needs a lot of H200 GPUs, hours to train, data preparation, and ongoing maintenance.
- **Existing model advantages:** Immediate deployment, proven performance, continuous updates.
- **Return on investment timeline:** Immediate vs 6-18 months for custom solutions.

AI agents work while you sleep

AI agents can be integrated with the software development lifecycle (SDLC) at each stage of the development phase—from coding and reviewing to testing, maintaining, and debugging. These AI agents can raise tickets, look for the issue, fix the issue, and create pull requests for final review. The cycle continues even after developers log off, as AI agents take over the system. By the next morning, developers arrive to find resolved tickets ready for final review.

The six-agent autonomous system we use is:

- **Boss agent:** Orchestrates the entire workflow and updates Jira status.
- **Agent 1:** Understands and triages Jira issues using standardised templates.
- **Agent 2:** Integrates with Grafana for root cause analysis.
- **Agent 3:** Communicates with stakeholders and provides RCA reports.
- **Agents 4 and 5:** Check out repositories and collaboratively debug (they literally argue with each other in the console logs like human developers).
- **Agent 6:** Creates pull requests and manages deployment.

Reasoning models vs fast inference

The most strategic decision is to understand when to use reasoning models versus fast inference. If you ask GPT-4.0 what '1+1' is, it will instantly respond with the answer '2'.

When asked the same question in the reasoning model, it will take significantly more time, engaging in philosophical contemplation about whether this was a tricky question. Reasoning models are 'slow thinkers'; reserve them for complex workflows and architectural decisions.

Challenges and solutions

Before using generative AI, companies should learn how to get the best results from AI, as it can give inaccurate code or data when working with large models. It's important to manage risks and ensure quality, so starting with small groups and analysing AI's benefits works best. Though AI costs money, it often saves more by speeding up work compared to hiring many new workers. Here's how we can manage risks, make cultural changes, and balance costs with benefits.

Managing risks and quality:

Generative AI carries real risks. Large models can produce inaccurate code or invent data. Even with expensive context windows, enterprise codebases can overwhelm an AI model. By using retrieval-augmented generation, which separates prompt engineering from text generation, we keep human oversight on customer-facing features. Business-related risks like developer resistance, customer concerns, and security are addressed with pilot programs, clear demonstrations of value, and strong vendor contracts that include on-premises options and audit trails.

Cultural transformation: Managing cultural change is as important as learning the technology. It is not wise to think that an organisation can simply deploy tools like Cursor and expect developers to immediately go ahead and start writing code with it. For us, this was a journey. Developers shifted from writing code line-by-line to overseeing AI-generated output, but they had to uncover a lot of challenges and build new habits before they reached this state. We spent a significant amount of time with our developers to train them on how to use AI to write code, and how to explain requirements to the AI in the right way—a practice we call 'context engineering'. In short, this dedicated training and change management are fundamental.

Costs versus benefits: Despite the scale, the cost can be surprisingly modest. Daily token consumption sounds high until you compare it with hiring dozens of additional developers. The annual cost of the entire AI platform is less than that of two senior engineers. Expenses include enterprise licences, secure infrastructure, training, and quality control, but the gains in delivery speed and business agility are far greater when implemented wisely.

Using AI well means knowing where it fits best — fast models for quick output, reasoning models for complex decisions, and pairing them with the right tools, agents, and workflows. But speed alone isn't enough. Success depends on managing risks, keeping human oversight where needed, easing cultural shifts, and measuring return on investment. When all these pieces work together, AI becomes not just a coding tool, but a competitive advantage. **END** 🐧

This article is based on the session titled 'Prompt, Set, Code!' delivered by Jaydeep Chakrabarty, Director of AI in Tech, and his team from Piramal Finance, at AI DevCon 25 in Bengaluru.

Transcribed and curated by: Janarthana Krishna Venkatesan, ex-journalist at the EFY Group.